Temporal trends and disparities of population attributable fractions of modifiable risk factors for dementia in China: a time-series study of the China health and retirement longitudinal study (2011–2018)

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Summary

Background In China, dementia poses a significant public health challenge, exacerbated by an ageing population and lifestyle changes. This study assesses the temporal trends and disparities in the population-attributable fractions (PAFs) of modifiable risk factors (MRFs) for new-onset dementia from 2011 to 2018.

Methods We used data from the China Health and Retirement Longitudinal Study (CHARLS), covering 75,214 person-waves. We calculated PAFs for 12 MRFs identified by the Lancet Commission (including six early-to mid-life factors and six late-life factors). We also determined the individual weighted PAFs (IW-PAFs) for each risk factor. Subgroup analyses were conducted by sex, socio-economic status (SES), and geographic location.

Findings The overall PAF for dementia MRFs had a slight increase from 45.36% in 2011 to 52.46% in 2018, yet this change wasn't statistically significant. During 2011–2018, the most contributing modifiable risk was low education (average IW-PAF 11.3%), followed by depression, hypertension, smoking, and physical inactivity. Over the eight-year period, IW-PAFs for risk factors like low education, hypertension, hearing loss, smoking, and air pollution showed decreasing trends, while others increased, but none of these changes were statistically significant. Sex-specific analysis revealed higher IW-PAFs for traumatic brain injury (TBI), social isolation, and depression in women, and for alcohol and smoking in men. The decline in IW-PAF for men's hearing loss were significant. Lower-income individuals had higher overall MRF PAFs, largely due to later-life factors like depression. Early-life factors, such as TBI and low education, also contributed to SES disparities. Rural areas reported higher overall MRF PAFs, driven by factors like depression, low education, and hearing loss. The study also found that the gap in MRF PAFs across different SES groups or regions either remained constant or increased over the study period.

Interpretation The study reveals a slight but non-significant increase in dementia's MRF PAF in China, underscoring the persistent relevance of these risk factors. The findings highlight the need for targeted public health strategies, considering the demographic and regional differences, to effectively tackle and reduce dementia risk in China's diverse population.

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Research in context

Evidence before this study

We conducted a literature search in PubMed and Web of Science for papers published before 10 August 2023, using the terms '("attributable fraction" OR "prevent*") AND ("dementia")'. The search terms were restricted to title and abstract. There were no language restrictions. The Lancet Commission on Dementia Prevention, Intervention, and Care (The Lancet Commission) used population attributable fractions (PAF) to estimate that 40% of new onset dementia cases worldwide were due to 12 potentially modifiable risks. These risks were obesity, physical inactivity, smoking, low education, diabetes mellitus, hypertension, depression, hearing impairment, alcohol consumption, social isolation, traumatic brain injury and air pollution. Prior to this research, some studies had explored the association between modifiable risk factors (MRFs) and dementia in China. However, specific insights into the temporal trends and disparities in PAF of these risk factors for new-onset dementia in China were less explored. Moreover, China lacks data on modifiable dementia risk stratified by sex, socio-economic status (SES), and geographic location, despite evident disparities.

Added value of this study

Our study stands as potentially the first to delineate the temporal trend of PAFs for modifiable risk factors for new-onset dementia in China. Spanning eight years, this longitudinal analysis offers a comprehensive exploration of how dementia risk factors have evolved, also serving as a measure for the impact of historical public health efforts. The study further delves into the disparities in dementia risk based on sex, SES, and geographic location, shedding light on how each risk factor distinctly contributes to these variations. Notably, our analysis indicates a moderate, yet not statistically significant, rise in the overall MRF PAF for new-onset dementia in China from 2011 to 2018. Additionally, our findings uncover variances in PAFs among different income levels and between rural and urban settings, pointing to the need for targeted intervention strategies.

Implications of all the available evidence

Our findings underscore the importance of addressing modifiable risk factors for new-onset dementia as integral to public health strategies in China and beyond. The research highlights that risk factors such as low education, depression, hypertension, smoking, and physical inactivity continue to be significant causes of new-onset dementia in China. These factors exhibit varying impacts based on sex, SES, and geographic location. This varying underlines the need for prevention strategies that are tailored to the demographic nuances. Moreover, the temporal trends observed in our study emphasize the necessity for ongoing monitoring and dynamic evaluation of intervention effectiveness, particularly in light of China's rapidly ageing population and urbanization trends. This approach not only aids in understanding the evolving landscape of dementia incidence but also in addressing persistent health disparities exacerbated by societal and environmental changes.

Introduction

The rising prevalence and economic impact of Alzheimer's disease and other dementias present a significant health and financial challenge in China. In 2019, dementia-related mortality in China rose to 22.5 per 100,000 from lower rates in 1990, making it the 5th leading cause of death.[1] The annual costs of dementia soared from 0.9 billion USD in 1990 to an estimated 114.2 billion USD by 2030.[2] Furthermore, the number of Chinese aged 60 and over with dementia is expected to triple to 48.98 million by 2050,[3] from 15.07 million in 2020.[4] All these trends underscore the urgent need to focus on modifiable risk factors (MRFs) as a key prevention strategy, especially in the absence of disease-modifying drugs.[5]

The concept of Population Attributable Fraction (PAF) plays a vital role in the research aimed at preventing diseases. It measures the percentage of cases that might be prevented if certain risk factors were eliminated.[6–9] The calculation of the PAF needs three parameters: the prevalence of risk factors (how common they are), their relative risk (RR) in relation to dementia (how strongly connected they are to dementia), and the communality between these factors (how much they overlap or share in common). The Lancet Commission on Dementia Prevention, Intervention, and Care ("the Lancet Commission") applied the PAF and estimated that globally, 40% of new-onset dementia cases could be linked to 12 MRFs. These include one early-life factor (low education), five mid-life factors (hearing loss, hypertension, obesity, excessive alcohol consumption, and traumatic brain injury [TBI]), and six later-life factors (smoking, depression, physical inactivity, social isolation, diabetes mellitus, and air pollution).[10] The dementia-related PAF associated with these MRFs has been calculated for various countries, including but not limited to Australia,[9,11] Canada,[12] Chile,[13] Greek,[14] Italy,[15] India,[7] Japan,[16] several nations in Latin America,[7] New

Zealand,[8] Sweden[17] and the United States.[6,14] These studies show considerable variance in their PAF estimates for MRFs, highlighting the importance of conducting evaluations that are tailored to specific countries and formulating corresponding intervention approaches.

The Chinese government is initiating a nationwide campaign to mitigate the impact of dementia. This initiative is highlighted in its 14th Five-Year Plan on Healthy Ageing, a collaborative effort from 15 departments, including the National Health Commission.[18] The plan emphasizes the importance of preventing and managing dementia, aiming to reduce and postpone its onset among older adults. A key component of this strategy is the formulation of a "National Action Plan on Dementia," which focuses on establishing a comprehensive framework for early screening, diagnosis, and intervention.

Despite recent progress, there remains a gap in our understanding of the PAF for new-onset dementia in China. Existing estimates of the China-specific PAF, primarily derived from analyses of 7–10 MRFs, were predominantly based on data preceding 2012[7,19] or are limited to a single-province dataset.[20] Crucially, these prior estimations provide previous estimation offers only a cross-sectional view.[7,19–21] Given the evolving epidemiological landscape in China, it is necessary to also examine the temporal trends of PAF to inform policy development and anticipate future challenges. Another significant gap in the current research is the lack of stratified PAF reports for dementia in China, considering variations of incidents of dementia in sex, socio-economic status (SES), and geographical differences (urban vs rural, provincial variations).[4,22–26] Understanding the primary contributors to new-onset dementia within these specific sub-groups is also essential for devising targeted and equitable policies with maximum health benefits.

Therefore, the focus of our research on the PAF of MRFs for new-onset dementia in China is driven by the urgent need to address the growing public health challenge posed by dementia, particularly in ageing populations.[3] This approach is pivotal for identifying preventive strategies that can significantly reduce the burden of dementia. By elucidating the potential impact of modifying risk factors, our study aims to contribute to the development of effective public health interventions and prevention approaches. The implications of our research are far-reaching, offering insights that could guide policymakers and healthcare professionals in implementing targeted measures to curb the incidence of dementia, ultimately enhancing the well-being and quality of life of millions.

Specifically, in this study, we utilized longitudinal data from China, covering the period from 2011 to 2018, to answer four questions: (1) What is the proportion of new-onset dementia cases in China linked to the 12 MRFs? (2) What are the primary MRFs driving the new-onset dementia in China? (3) How has the proportion of new-onset dementia cases associated with the 12 MRFS evolved in China, and in what manner has the contribution of each factor varied within this timeframe? and (4) Do these trends vary across different demographics, including sex, SES, and geographic location?

Methods

Data source and participants

This study utilized data from the China Health and Retirement Longitudinal Study (CHARLS), a publicly accessible dataset in China.[27] CHARLS is a longitudinal survey that conducts biennial follow-ups with participants. It aims to represent individuals aged 45 and above residing in private households across mainland China. In brief, CHARLS employs a multi-stage probability sampling design to ensure national representativeness.[27] The sampling frame covers 28 provinces, 150 counties/districts, and 450 villages/urban communities.[27,28] Comparisons with the 2010 Chinese Census data show that CHARLS is largely representative of the Chinese population aged 45+ in terms of age, gender, and health risk behaviour. For example, in baseline, the proportion of females aged 45–65 in CHARLS is 51.6%, compared to 49.1% in the census. CHARLS has maintained high response rates across its survey waves, with an overall response rate of 80.5% in the 2011 baseline survey and rates of 82.6%, 82.1%, and 83.8% for 2013, 2015, and 2018 waves, respectively.[27,29] To maintain representativeness over time, CHARLS employs several strategies. These include the use of weighting to adjust for any non-response or sampling biases, and periodically refreshing its cohort in each wave to maintain age representation.[27] The weights account for the differential probabilities of selection, non-response, and post-stratification to known population totals. Here's a more detailed explanation of how the weights are calculated in CHARLS.[27,29] Supplementary Table S1 shows the survey participation from 2011 to 2018, detailing the number of participants retained from previous survey years and the number of new participants added in each survey year.

Data collection was performed by trained interviewers. In cases where respondents were either unable or unwilling to participate, proxy respondents, typically a spouse or another family member, were employed for data collection. These proxy interviews comprise approximately 8% of all interviews.[27] Data collected by CHARLS included socio-demographic factors, health status and diagnoses, health behaviours, social networks, health examination results, and blood samples. The methodology, sampling procedures, and quality assurance measures of CHARLS have been thoroughly documented in other publications.[27]

The data are publicly available. The use of secondary de-identified data made this study exempt from the institutional review board. CHARLS was approved by the Ethical Review Committee of Peking University (IRB00001052-11015).

For our analysis, we utilized data from wave 1 to wave 4 of CHARLS, covering the period from 2011 to 2018. Wave 5 (conducted in 2020) was excluded from this study due to the impact of COVID-19-related measures on 12 MRFs like depression, physical activity, social isolation, and alcohol assumption.[30–33] The sample sizes for each wave ranged between 17,708 person-wave and 19,816 person-wave. The estimation of risk factor prevalence may be distorted by the inclusion of individuals diagnosed with dementia, especially considering that certain risk factors, such as obesity and depression, are likely to decrease or increase after the onset of dementia.[34–36] Our analysis included all participants except those who had previously been diagnosed with dementia or related memory disorders, or exhibited probable dementia at the time of their initial enrolment in the CHARLS. Probable dementia was considered to address the issue of dementia's potential under-diagnosis.[37,38] Diagnosed dementia or related memory disorders was determined by the self-reported question, "Have you been diagnosed with memory-related disease (such as dementia, brain atrophy, and Parkinson's disease) by a doctor". Probable dementia cases were judged using a cognitive assessment with a total score of 25 points, encompassing tests for immediate and delayed word recall (with a maximum score of 10) and self-evaluated memory abilities (with a maximum score of 5).[39,40] Participants who scored 1.5 standard deviations below the average cognitive score, with adjustments made for their education level, were categorized as likely having dementia.[39,40]

Modifiable risk factors

We integrated all the 12 factors identified by the Lancet Commission as crucial for evaluating dementia risk. The comprehensive justification for selecting these factors can be found in the Lancet Commission's reports.[10,41] We endeavoured to align with the definitions of risk factors as specified by the Lancet Commission, wherever possible. When the CHARLS dataset lacked corresponding data, we resorted to using definitions from other studies,[6–9,17] that also calculated PAF in line with the Lancet Commission's framework. A detailed definition of each risk factor is presented in Table 1.

For our subgroup analysis, sex was classified as participants' self-reported sex at birth (male vs female). SES was estimated in each wave using a composite measure of non-housing financial wealth, which includes income sources such as earnings, savings, social benefits, and investments like stocks and bonds. The geographic location of participants for each wave was determined through the interview process conducted by trained personnel. They documented the specific area where participants resided during each survey phase.

Calculation of population attributable fractions

The approach for calculating the PAF adheres to established protocols documented in various sources.[7,10,11,42] In brief, calculating PAF requires three key elements: the prevalence rate of the risk factors, the strength of their association with dementia (measured through relative risk, RR), and the overlap or communality between these risk factors. In this study, the result, PAF, is presented as a percentage (PAF%).

$$Individual\ PAF_i = \frac{prevalence_i \times (RR_i - 1)}{1 + prevalence_i \times (RR_i - 1)},\quad i = 1, 2 \ldots 12 \quad \text{Formula 1}$$

In which, i indicates corresponding parameters or estimates belonging to risk factor i.

$$Overall\ PAF = 1 - (1 - weight_1 \times PAF_1) \times (1 - weight_2 \times PAF_2) \ldots (1 - weight_{12} \times PAF_{12}) \quad \text{Formula 2}$$

In which,

$$Weight_i = 1 - community_i \quad \text{Formula 3}$$

$$Individual\ weighted\ PAF_i = \frac{Individual\ PAF_i}{\sum_1^{12} Individual\ PAF_i} \times Overall\ PAF \quad \text{Formula 4}$$

Estimation of prevalence

Acknowledging that certain risk factors like obesity may decrease with the onset of dementia, the Lancet Commission suggests considering risk factors within specific life stages.[10] Our study estimated the prevalence of these risk factors among CHARLS participants based on the age ranges recommended by the Lancet Commission.[10] As detailed in Table 1, the prevalences of hearing loss, hypertension, obesity, excessive alcohol consumption, and TBI were calculated among individuals aged 45–64, while the prevalences of smoking, physical inactivity, social isolation, diabetes mellitus, and air pollution were calculated among individuals aged 65 and above. Low education was treated as an early-life factor, and the Lancet Commission suggested calculating its prevalence among those aged <45. However, as CHARLS only includes individuals over 45 and educational levels in adults typically remain stable and do not change with disease progression, we used the prevalence of low education among individuals aged 45–64 as a proxy for the prevalence of low education among those aged <45. Notably, the assessment of air pollution exposure was conducted through the integration of CHARLS data with information from

Risk factors	Definition in the report of the Lancet commission	Definition in this study
Early-life (age <45 years)		
Low education (≤65)	None or primary school only; or no secondary school education; or formal education to a maximum age of 11–12 years.	Self-reported highest education attained is less than lower secondary.
Midlife (age 45–64 years)		
Hearing loss	Hearing loss is present at a threshold of 25 dB, which is the World Health Organization threshold for hearing loss.	Meet at least one of the following criteria: 1. Self-rated hearing ability as poor, regardless of the use of hearing aids; 2. Having hearing disabilities
Hypertension	Systolic BP of 130 mm Hg or higher.	Hypertension from blood pressure (i.e., systolic ≥130 mmHg or diastolic ≥90 mmHg) or self-reported diagnosis of hypertension.
Obesity	BMI ≥ 30	BMI ≥ 30
Excessive alcohol	Drinking more than 21 units (168 g) of alcohol weekly.	Self-reported drinking frequency of more than half a week, including 4–6 days a week, daily, twice a day, or more than twice a day.
Traumatic brain injury (TBI)	The International Classification of Disease (ICD) defines mild TBI as concussion and severe TBI as skull fracture, oedema, brain injury or bleeding.	Meet both of the following criteria: 1. Ever experienced a serious traffic accident or injury, 2. Feel pain in the head or have a speech impediment
Later-life (age ≥65 years)		
Smoking	No definition provided	Self-reported question on current smoking habit
Depression	No definition provided	CESD-10 ≥ 10, or self-reported diagnosis of emotional, nervous, or psychiatric problems.
Physical inactivity	No definition was provided. The report summarised that at least weekly midlife moderate-to-vigorous physical activity (breaking into a sweat) was associated with reduced dementia risk over 25 years of follow-up, and that more than 2.5 h of self-reported moderate-to-vigorous physical activity per week lowered dementia risk over 10 years.	Self-reported at least 3 days per week taking part in sports or activities that are vigorous or moderate which lasted for at least 10 min at a time.
Social isolation	Social isolation was not measured directly. Instead, living alone (i.e., no cohabitation) was used as a proxy measure.	Living alone, no other people living in the household.
Diabetes mellitus	No definition provided	HbA1c ≥ 6.5% or self-reported diabetes mellitus.
Air pollution	No definition was provided. The report summarised that high nitrogen dioxide (NO2) concentration (>41.5 μg/m^3; adjusted HR 1.2, 95% CI 1.0–1.3), fine ambient particulate matter (PM)2.5 from traffic exhaust (1.1, 1.0–1.2) and PM2.5 from residential wood burning (HR = 1.6, 95% CI 1.0–2.4 for a 1 μg/m^3 increase) are associated with increased dementia incidence.	Poor air quality: daily PM2.5 ≥ 35 μg/m^3 Air pollution days in a year are in the bottom 20% quantile of annual air pollution days during follow-up nationwide.

Table 1: **Definitions of dementia risk factors.**

ChinaHighAirPollutants, aligning with the survey communities' locations. ChinaHighAirPollutants database provided daily measurements of PM2.5 levels.[43] A community was classified as having experienced significant air pollution if the number of days with poor air quality in a survey year fell within the lowest 20% of all recorded annual air pollution days nationwide during the follow-up period. We defined a day as having poor air quality if the daily PM2.5 concentration exceeded 35 μg/$\hat{m}^3$, a threshold recognized as unhealthy for the population.[44]

Estimates of relative risk

The estimates of the relative risk (RR) of the association of the individual MRFs with dementia were extracted from the 2020 Lancet Commission Report,[10] as detailed in Supplementary Table S2. This report utilized a systematic review and meta-analyses to obtain RR values for the twelve MRFs.

Communality

Individuals often have multiple concurrent risk factors, like hypertension and diabetes. To calculate the PAF for new-onset dementia, it's important to adjust for the overlap among these factors, known as communality.[42] This study calculates communality using established dementia PAF research methods: first, establish a tetrachoric correlation matrix of risk factors for principal components analysis, and determine communality from the sum of squared loadings of principal components with an eigenvalue above 1.[7,10,11,42] Details of the communality calculation can be found in Appendix S1.

Data analysis

All analyses were carried out using R project software (version 4.3.0). Sampling weights from CHARLS, reflecting the complex sampling design, were applied to produce accurate estimates. Statistical significance was assessed using two-tailed p-values, with $p < 0.05$ as the threshold.

Missing values for the study variables ranged from 0.003% (for sex) to 23.5% (for wealth status). We have implemented 25 multiple imputations by chained equations to minimize potential biases arising from missing data.[45] The imputation process included covariates such as the 12 MRFs, socio-demographic factors (age, sex, wealth status, and living place), and sampling weights. Sampling weights were included to account for potential unequal sampling probabilities of participants suggested by previous study.[46]

To address uncertainty in PAF estimation, we employed Monte Carlo simulations (n = 1,000,000), using beta distributions for prevalence and normal distributions for the logarithm of RRs.[13] We reported the mean, 2.5%, and 97.5% quantiles from the PAF distributions.

To examine the temporal trend (average percentage change [APC]) per year of the PAF, linear regressions were conducted with the continuous form of the year as the predictor and PAF as the outcome.

All the above analyses were repeated by sex, SES, and geographic location. We extracted the between-group variation of PAF across these sub-groups by life stage or by risk factors, from the ANOVA test. This between-group variance enables us to understand which life-stage or which risk factor disproportionately affects overall PAF disparity across these groups.

To further validate the stability of our PAF estimates, we conducted two sensitivity analyses. Firstly, we calculated relative risks (RRs) using data from the CHARLS dataset, rather than relying on the values extracted from the Lancet Commission Report (for details on the calculation, please refer to Appendix S2). Secondly, we repeated the analyses without imputation by excluding cases with missing values, to evaluate the impact of missing data on our findings.

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the article.

Results

Description of the total data set

The CHARSL from 2011 to 2018 encompassed 75,214 person-waves. The participants had an average age of 59.5 years (SD 10.4), and females constituted 52.4% of the sample. Over the study period, the predominant risk factor was low education, present in 87.7% of participants. This was followed by exposure to air pollution (79.5%), hypertension (44.9%), depression (32.4%), physical inactivity (29.6%), smoking (27.3%), TBI (17.2%), hearing loss (16.1%), excessive alcohol consumption (13.7%), diabetes (11.0%), obesity (6.6%), and social isolation (6.1%). The basic description of the socio-demographic factors and 12 modifiable risk factors by survey year was presented in Table 2. With CHARLS surveying going, more people aged 65 or over ($p < 0.001$), but no significant difference in sex ($p = 0.318$). The distribution of missing values by survey year was presented in Supplementary Table S3. Participants with the lowest wealth status, living in rural areas, and having low education had greater proportions of missing values.

PAF trajectory in China

Between 2011 and 2018, the overall MRF PAF for new-onset dementia rose from 45.36% (95% confidence interval [CI] [27.02, 59.85]) in 2011 to 52.46% (95% CI [32.02, 67.67]) in 2018 (Fig. 1, Panel C). However, this increase was not statistically significant (APC 0.99, 95% CI [−0.16, 2.13]). The overall MRF PAF was roughly evenly split between factors occurring in early-to mid-life and those in later life (Fig. 1, Panel A and B). During the study period, the increase in PAF due to early- and mid-life factors wasn't significant (APC 0.24, 95% CI [−0.92, 1.41]), whereas the rise attributable to later-life factors was significant (APC 0.74, 95% CI [0.32, 1.16]).

From 2011 to 2018, the analysis of individual weighted PAFs (IW-PAFs) for new-onset dementia by risk factors revealed that low education contributed the most, with an average IW-PAF of 11.3% (Fig. 2). This was followed by depression (7.61%), hypertension (6.43%), smoking (4.26%), physical inactivity (3.77%), TBI (3.76%), hearing loss (3.16%), air pollution (2.34%), social isolation (2.30%), diabetes mellitus (2.21%), obesity (1.69%), and excessive alcohol consumption (0.80%) (Fig. 2). Over the eight-year period, the IW-PAFs for low education, hypertension, hearing loss, smoking, and air pollution exhibited downward trends, whereas the remaining seven factors showed upward trends. However, none of these trends were statistically significant (Fig. 2).

PAF trajectory by sex

The analysis of the PAF by sex indicated similar levels of PAF in early- and mid-life factors, later-life factors, and all 12 MRFs for both females and males (Fig. 3). From 2011 to 2018, both sexes exhibited a non-significant increase in PAF across these categories (Fig. 3).

When analysed the IW-PAFs by sex (Fig. 4), females showed higher average IW-PAFs for TBI, social isolation, and depression, whereas males had higher average IW-PAFs for excessive alcohol consumption and smoking. During the eight-year period, IW-PAFs for hearing loss in males showed a significant downward trend (APC −0.08, 95%CI [−0.15, −0.01]), contrasting with the non-significant upward trend in females (APC 0.001, 95% CI [−0.13, 0.13]) (Fig. 4). The IW-PAFs for the remaining 11 MRFs displayed consistent but statistically non-significant trends in both sexes over time (Fig. 4).

PAF trajectory by socio-economic status

The assessment of the overall MRF PAF by SES indicated a higher PAF in lower-income groups compared to higher-income ones (Fig. 5). This difference was primarily due to the greater variance in PAF linked to later-life factors (between-group variance across SES

Variable	Year (=2011) (n = 16,207)	Year (=2013) (n = 18,356)	Year (=2015) (n = 20,878)	Year (=2018) (n = 19,773)	All (n = 75,214)	p
Socio-demographic factors						
Age (≥65)	3788 (23.4%)	5272 (28.7%)	6231 (29.8%)	7401 (37.4%)	22,692 (30.2%)	<0.001
Sex (=Female)	8410 (51.9%)	9610 (52.4%)	10,941 (52.4%)	10,456 (52.9%)	39,417 (52.4%)	0.318
Wealth status						
Lowest	3329 (20.5%)	3685 (20.1%)	4144 (19.8%)	4763 (24.1%)	15,921 (21.2%)	<0.001
2	3295 (20.3%)	3659 (19.9%)	4129 (19.8%)	3290 (16.6%)	14,373 (19.1%)	<0.001
3	3188 (19.7%)	3426 (18.7%)	4260 (20.4%)	3937 (19.9%)	14,811 (19.7%)	<0.001
4	2976 (18.4%)	3622 (19.7%)	3983 (19.1%)	3834 (19.4%)	14,415 (19.2%)	<0.001
Highest	3419 (21.1%)	3964 (21.6%)	4362 (20.9%)	3949 (20.0%)	15,694 (20.9%)	<0.001
Urban (=yes)	6650 (41.0%)	7376 (40.2%)	8465 (40.5%)	7976 (40.3%)	30,467 (40.5%)	0.408
Modifiable risk factors						
Low education (=yes)	14,207 (87.7%)	16,106 (87.7%)	18,345 (87.9%)	17,274 (87.4%)	65,932 (87.7%)	0.462
Hearing loss (=yes)	2619 (16.2%)	2933 (16.0%)	3152 (15.1%)	3430 (17.3%)	12,134 (16.1%)	<0.001
Hypertension (=yes)	7527 (46.4%)	8784 (47.9%)	9935 (47.6%)	7561 (38.2%)	33,807 (44.9%)	<0.001
Obesity (=yes)	936 (5.8%)	1210 (6.6%)	1339 (6.4%)	1515 (7.7%)	5000 (6.6%)	<0.001
Excessive alcohol (=yes)	1764 (10.9%)	2864 (15.6%)	2934 (14.1%)	2752 (13.9%)	10,314 (13.7%)	<0.001
Traumatic brain injury (=yes)	2370 (14.6%)	1874 (10.2%)	3267 (15.6%)	5404 (27.3%)	12,915 (17.2%)	<0.001
Smoking (=yes)	4807 (29.7%)	4586 (25.0%)	5844 (28.0%)	5278 (26.7%)	20,515 (27.3%)	<0.001
Depression (=yes)	5318 (32.8%)	5413 (29.5%)	6593 (31.6%)	7051 (35.7%)	24,375 (32.4%)	<0.001
Physical inactivity (=yes)	2923 (18.0%)	3028 (16.5%)	4967 (23.8%)	11,363 (57.5%)	22,281 (29.6%)	<0.001
Social isolation (=yes)	869 (5.4%)	885 (4.8%)	1059 (5.1%)	1777 (9.0%)	4590 (6.1%)	<0.001
Diabetes mellitus (=yes)	1214 (7.5%)	1633 (8.9%)	2922 (14.0%)	2533 (12.8%)	8302 (11.0%)	<0.001
Air pollution (=yes)	14,797 (91.3%)	16,791 (91.5%)	16,431 (78.7%)	11,782 (59.6%)	59,801 (79.5%)	<0.001

For socio-demographic factors and unweighted modifiable risk factors, data presented as the number (percentage), and p-value was extracted from the Chi-squared test.

Table 2: **Basic description of socio-demographic factors and 12 modifiable risk factors by survey year.**

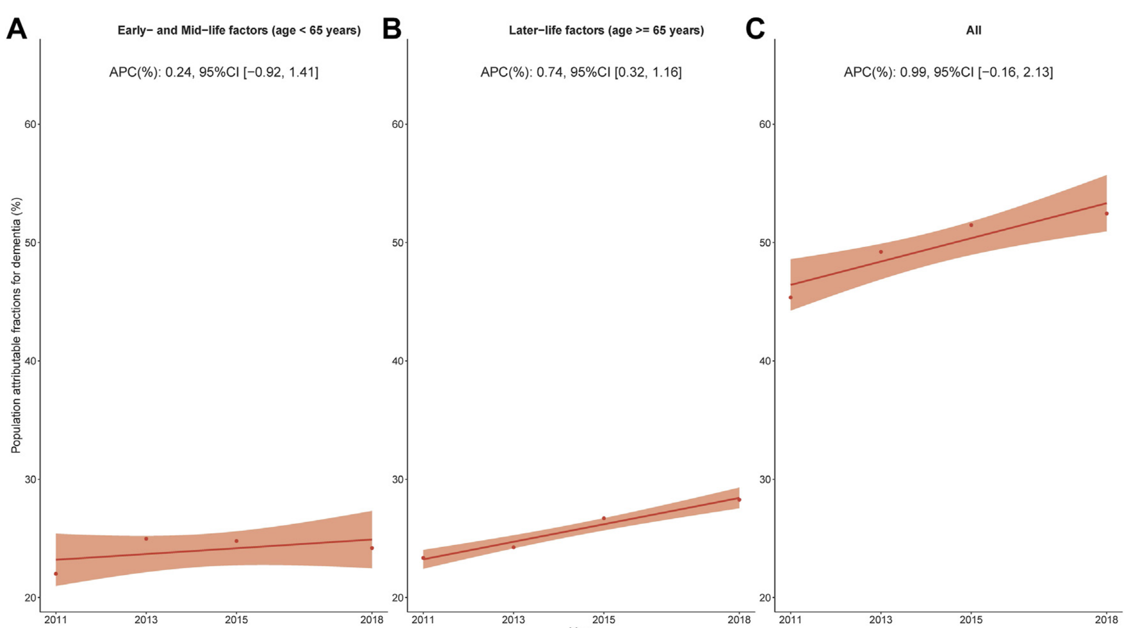

Fig. 1: **Overall temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year.

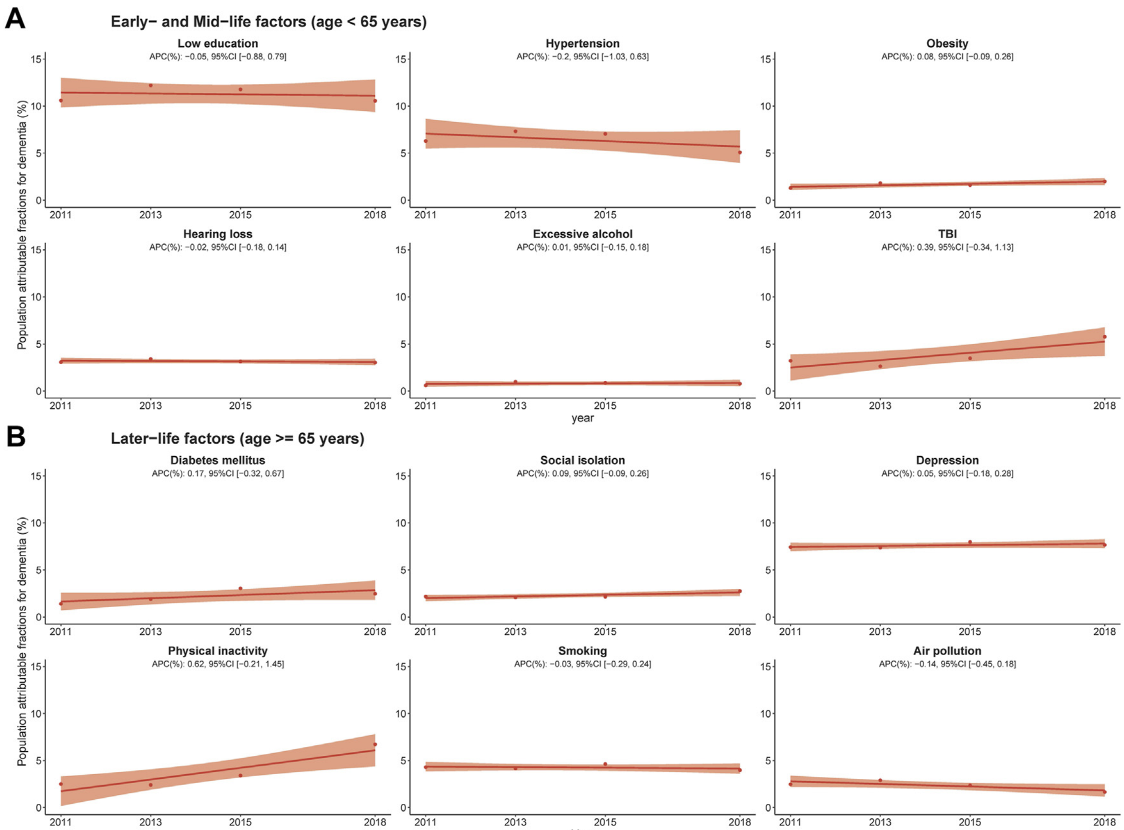

Fig. 2: **Temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018, by risk factor.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year. The same figure but with the range of the Y-axis adjusted accordingly based on the observations for each factor, can be found in Supplementary Fig. S1.

groups = 46.8 in later-life factors vs 10.7 in early- and mid-life factors) (Fig. 5, Panel A and B). Between 2011 and 2018, all SES groups showed increases in PAF for early- and mid-life factors, later-life factors, and all the 12 MRFs, except the non-significant decreasing trend in the highest SES group for early- and mid-life factors (Fig. 5).

In terms of IW-PAF by SES, the primary contributors to disparities were depression (between-group variance of IW-PAF across SES = 11.0), TBI (3.91), social isolation (2.84), low education (2.63), and hearing loss (2.6) (Fig. 6). Over the eight-year period, most IW-PAFs of risk factors showed consistent, non-significant trends across different SES groups (Fig. 6). However, obesity's IW-PAF significantly increased in the lowest SES group (APC 0.02, 95% CI [0.02, 0.02]), diabetes exhibited a significant upward trend in the highest SES group (APC 0.25, 95% CI [0.01, 0.49]), and social isolation exhibited a significant upward trend in the lowest SES group (APC 0.44, 95% CI [0.11, 0.78]) (Fig. 6).

PAF trajectory by geographic location

The analysis of PAF in rural vs urban areas showed that rural regions had a higher overall MRF PAF than urban areas (Fig. 7, Panel C). This disparity was largely driven by the higher PAF in later-life factors in rural areas (between-group variance of PAF across rural-urban = 44.2 for later-life factors vs 11.6 for early- and mid-life factors) (Fig. 7, Panel A and B). From 2011 to 2018, both rural and urban areas experienced an increase in overall MRF PAF, but this increase was statistically significant only in rural areas (APC 1.44, 95% CI [0.1, 2.78]) (Fig. 7, Panel C). The trend in PAF related to later-life factors reflected this overall pattern (Fig. 7, Panel B). For early- and mid-life factors, both rural and urban areas showed a non-significant increasing trend (Fig. 7, Panel A). These varying trends led to a widening gap in overall MRF PAF between rural and urban areas, increasing from 6.02 in 2011 to 9.27 in 2018.

When examining IW-PAFs by rural-urban settings (Fig. 8), rural areas had higher average IW-PAFs for factors like low education, hearing loss, TBI, depression,

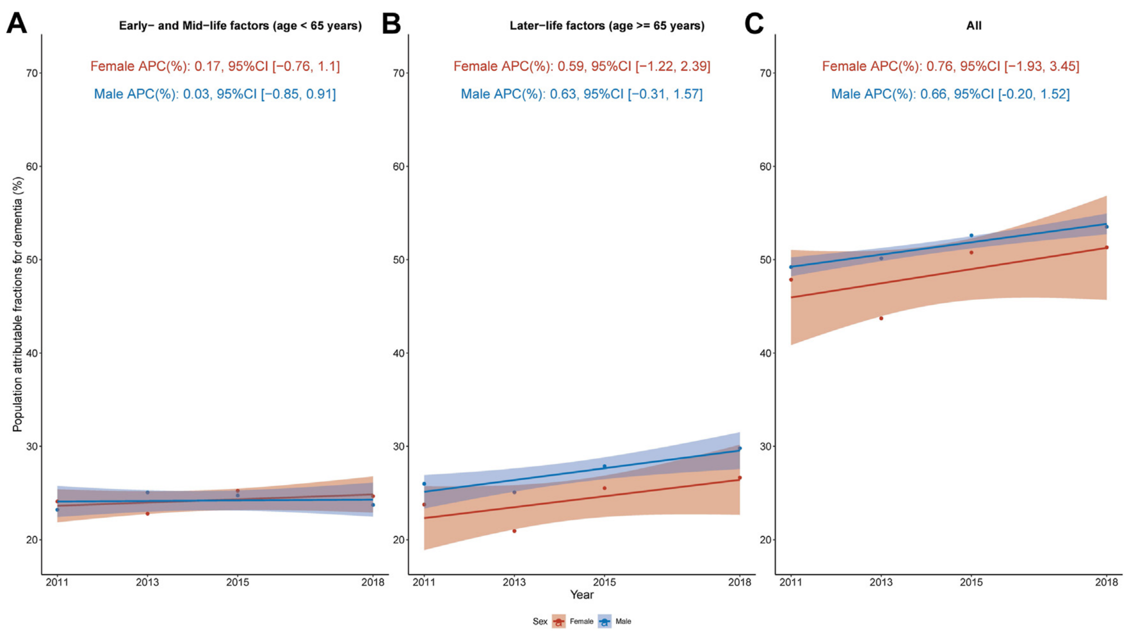

Fig. 3: **Temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018, by sex.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year.

and smoking. In contrast, urban areas had higher average IW-PAFs for obesity and diabetes mellitus (Fig. 8). The primary factors contributing to the rural-urban disparity were depression (between-group variance of IW-PAF across rural-urban = 14.5), low education (7.13), hearing loss (4.48), TBI (4.32), smoking (2.31), and diabetes mellitus (1.82) (Fig. 8). Over the study period, smoking's IW-PAF significantly decreased in urban (APC −0.02, 95% CI [−0.02, −0.01]), while the IW-PAFs for other 11 MRFs displayed stable, non-significant trends in both rural and urban settings.

Fig. 9 illustrates the provincial distribution and temporal evolution of the overall PAF in China from 2011 to 2018, with more detailed visualizations in Supplementary Figs. S7–S34. Regions in western China and eastern or coastal areas generally showed lower overall PAFs compared to central China. However, there was an increasing trend in overall PAF across most provinces, notably with significant rises in Hebei (APC 2.25, 95% CI [1.4, 3.1]), Hunan (APC 0.79, 95% CI [0.1, 1.48]), Qinghai (APC 2.91, 95% CI [0.65, 5.16]), and Sichuan (APC 1.4, 95% CI [0.61, 2.2]). Conversely, Beijing, Henan, Shanghai, and Tianjin exhibited declining trends, though not statistically significant. The gap between the lowest and highest overall MRF PAF by province increased from 25.5% in 2011 to 40.6% in 2018. The provincial distribution of PAFs related to early- and mid-life factors (Supplementary Fig. S5) and later-life factors (Supplementary Fig. S6), along with their detailed visualizations (Supplementary Figs. S7–S34), are also provided. Chongqing (APC 0.43, 95% CI [0.12, 0.74]), Guizhou (APC 1.15, 95% CI [0.17, 2.14]), Hebei (APC 0.68, 95% CI [0.29, 1.08]), and Jiangxi (APC 1.33, 95% CI [0.15, 2.5]) showed significant upward trends, while Henan (APC −0.5, 95% CI [−0.94, −0.06]) showed significant downward trend, in PAFs associated with early- and mid-life factors. Significant increases in PAFs for later-life factors were observed in Hebei (APC 1.57, 95% CI [0.74, 2.4]) and Qinghai (APC 1.66, 95% CI [0.63, 2.69]). The gap between the lowest and highest PAF for early- and mid-life factors by province increased from 11.3% in 2011 to 17.8% in 2018. The gap between the lowest and highest PAF for later-life factors by province increased from 14.2% in 2011 to 23.4% in 2018.

Supplementary Figs. S35–S62 present analyses of IW-PAFs across various provinces. Notably, IW-PAF for low education showed a significant decrease in Beijing but a significant increase in Jiangxi. IW-PAFs for depression experienced significant increases in Hebei. IW-PAFs for hypertension experienced significant decreases in Beijing. IW-PAFs for smoking had a significant rise in Chongqing, Guangdong, Hebei, Jiangxi, and Yunnan. IW-PAFs for TBI had significant upward trends in Guizhou, Jiangsu, and Qinghai. IW-PAFs for hearing loss had a significant rise in Xinjiang and Yunnan. IW-PAFs for air pollution had significant decrease in Inner Mongolia. IW-PAFs for social

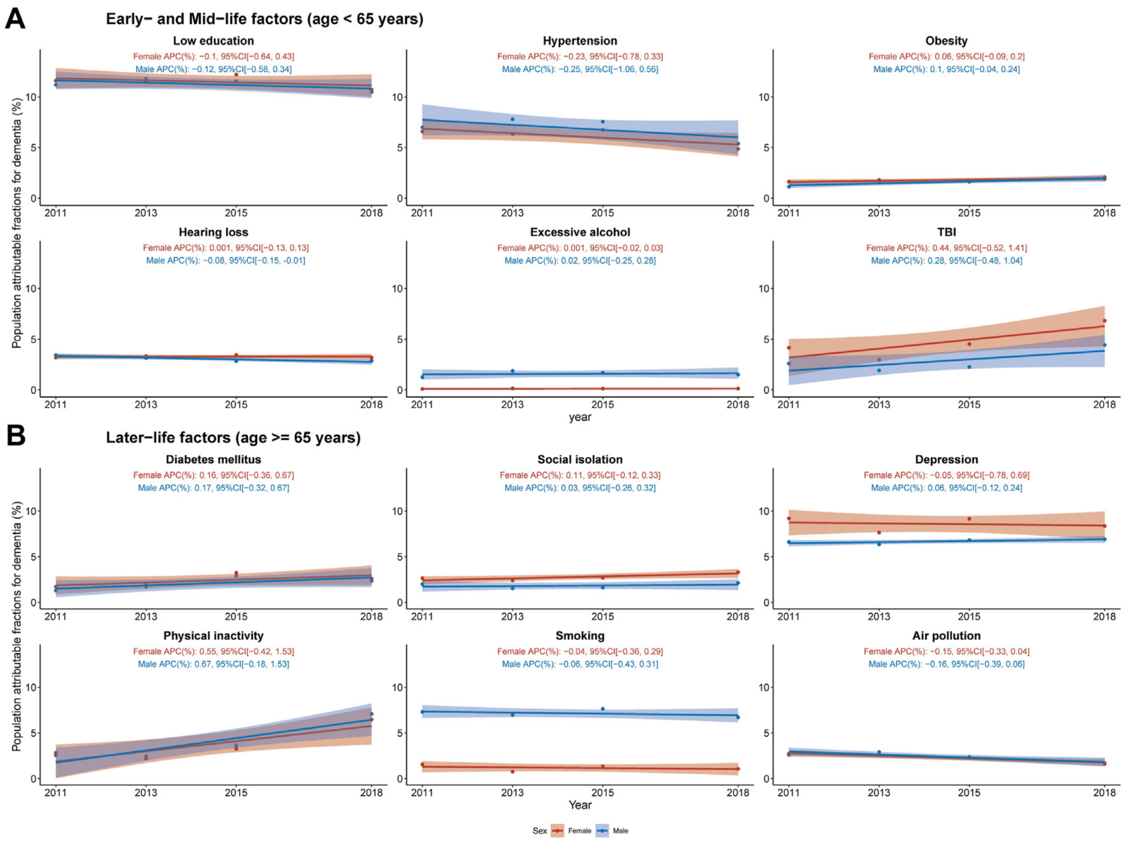

Fig. 4: **Temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018, by sex and risk factor.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year. The same figure but with the range of the Y-axis adjusted accordingly based on the observations for each factor, can be found in Supplementary Fig. S2.

isolation had significant increases in Hebei, Qinghai, Shandong, Shanxi, and Xinjiang. IW-PAFs for diabetes mellitus rose significantly in Chongqing, Hebei, and Shandong. IW-PAFs for obesity showed significant increases in Chongqing, Hebei, Sichuan, and Yunnan. IW-PAFs for excessive alcohol consumption significantly increased in Shanxi.

Sensitivity analyses, using RRs deprived from the CHARLS (Supplementary Figs. S63–S70), and repeated analyses without imputation (Supplementary Figs. S71–S78), confirmed our primary results, in terms of temporal trends and the relative importance of each risk factor.

Discussion

Statement of principal findings

In China, the overall MRF PAF for new-onset dementia increased slightly from 45.36% in 2011 to 52.46% in 2018, although this rise was not statistically significant. The IW-PAFs identified low education as the most substantial contributor to modifiable dementia risk, driving 11.3% of new-onset cases of dementia during 2011–2018, followed by depression (7.61%), hypertension (6.43%), smoking (4.26%), physical inactivity (3.77%), and other factors. Over the eight-year period, IW-PAFs for risk factors like low education, hypertension, hearing loss, smoking, and air pollution showed decreasing trends, while others increased. When disaggregated by sex, both PAF levels and trends were similar across sexes, but women exhibited higher IW-PAFs for TBI, social isolation, and depression, whereas men had higher IW-PAFs for excessive alcohol consumption and smoking. Notably, among 12 MRFs, IW-PAF for hearing loss in men showed a significant downward trend. Lower-income groups displayed a higher overall MRF PAF than higher-income groups, primarily due to later-life factors, like depression and social isolation. Factors from early and middle life, such as low education, TBI, and hearing loss, were also identified as contributors to the variations in modifiable dementia risk based on SES. Rural areas had a higher

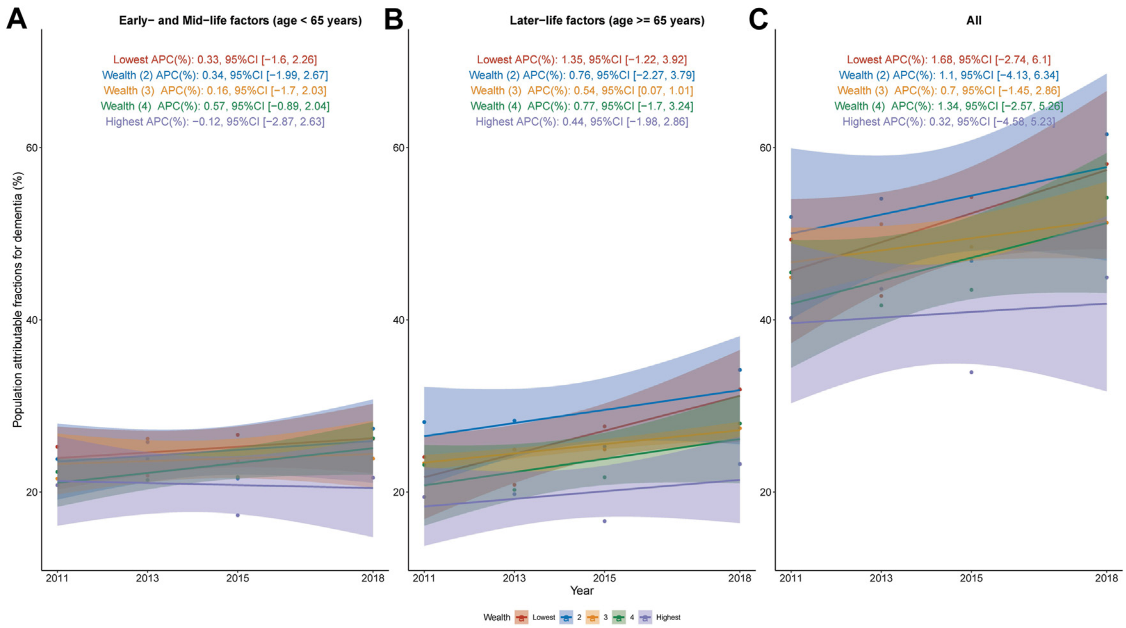

Fig. 5: **Temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018, by socio-economic status.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year.

overall MRF PAF compared to urban areas, driven mainly by later-life factors like depression, smoking, and diabetes. Additionally, low education, hearing loss, and TBI, categorized as early-to mid-life factors, also partially contributed to the observed rural-urban disparity. Geographic results by province were also presented. A critical examination of the temporal PAF trends stratified by sex, SES, and geographic location suggests that the identified gaps across rural-urban, or provinces have widened over time.

Interpretation

The slight increase in the MRF PAF for new-onset dementia suggests that the current approaches in China to managing modifiable dementia risk factors might not be fully effective. This finding aligns with recent data on dementia prevalence in China. Two large, multiregional studies found that the prevalence of dementia among individuals aged 65 years or older was 5.14% (95% CI 4.71–5.57) in 2014 and 5.60% (3.50–7.60) in 2019.[26,47] While the rise was not statistically significant, the absence of decline points to a need for re-evaluating existing health policies, enhancing effective ones and piloting new initiatives.

In recent years, China has implemented various policies aimed at reducing the burden of non-communicable diseases, including those related to dementia risk factors. For example, the “China National Plan for NCD Prevention and Treatment” and the “Healthy China 2030” plan. These efforts collectively set targets for reducing smoking prevalence, increasing physical activity, and improving the management of chronic conditions like hypertension and diabetes.[48,49] However, the impact of these policies on the specific risk factors for dementia may have been limited, as suggested by the lack of significant declines in their respective IW-PAFs. Several factors may contribute to the apparent ineffectiveness of these policies in reducing the burden of modifiable dementia risk factors. First, the complexity of the problem may require a more comprehensive and coordinated approach across multiple sectors, including health, education, and social services.[50] Second, the implementation of policies may be hindered by resource limitations, competing priorities, or inconsistencies across different regions of China.[51] Third, the time frame for seeing the impact of policies on dementia risk factors may be longer than the eight-year period covered in our study, as many of these factors have their roots in early life experiences and accumulate over the life course.[50] To improve the effectiveness of policies targeting modifiable dementia risk factors in China, we recommend: conducting a thorough review of existing policies; developing a comprehensive, multi-sectoral strategy; ensuring adequate resources and political support; prioritizing the reduction of health inequalities; and establishing robust monitoring and evaluation systems to inform ongoing improvements.

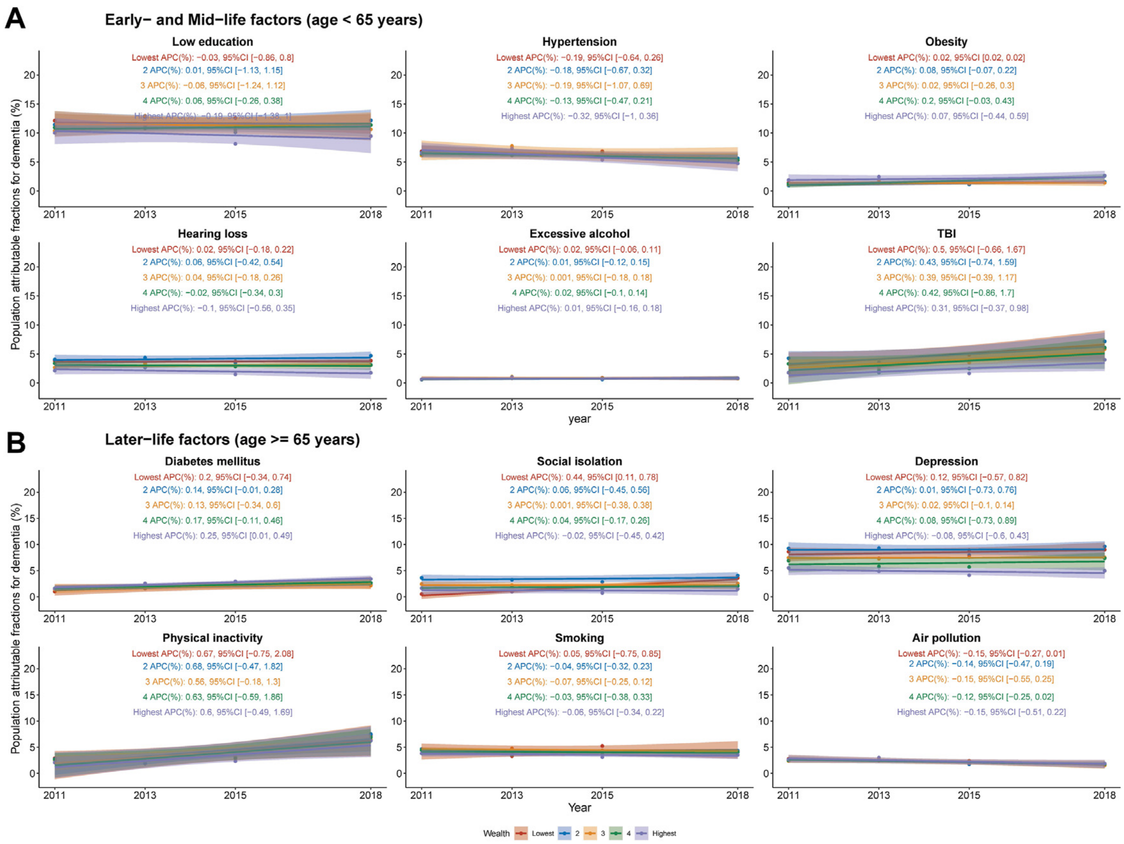

Fig. 6: **Temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018, by socio-economic status and risk factor.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year. The same figure but with the range of the Y-axis adjusted accordingly based on the observations for each factor, can be found in Supplementary Fig. S3.

The observed trend could also potentially be attributed to the rapidly ageing population, unintended consequences of economic development (like increasing left-behind populations), and urbanization-driven lifestyle changes (likes less physical activity), which may be offsetting the effects of current public interventions.[52–54] Comparatively, a study from Western countries exhibited levels of success in managing similar MRF for dementia, indicating that intervention effectiveness could be highly contingent on contextual factors such as cultural, socio-economic, and healthcare system contexts.[55] A global perspective reveals that China's overall PAF for dementia MRFs (around 50%) falls on the higher end of the spectrum–from Mozambique's 24% to Australia's 48%.[6–9,11–17,19–21] This variation might primarily reflect the difference in populations and the prevalence of risk factors across these regions. Nevertheless, China's positioning towards the higher end in PAF suggested significant room for improvement.

Our findings on the IW-PAFs of 12 MRFs offer crucial direction for dementia prevention in China. Low education played a significant role, accounting for 11.3% of new-onset dementia cases from 2011 to 2018. Depression, hypertension, smoking, and physical inactivity followed by low education in importance. These results emphasize the impact of social determinants and lifestyle on dementia. However, these findings contrast with findings in other countries[8,13] where metabolic factors like hypertension and obesity are more prominent. This inconsistency highlights the unique socio-economic and cultural influences on modifiable dementia risk factors in China. Although the reductions in IW-PAFs for low education, hypertension, hearing loss, smoking, and air pollution were not statistically significant, the findings hint at a slow impact of current health policies and interventions, emphasizing the need for sustained efforts. However, the increasing or stable trends in factors like depression, physical inactivity, TBI, diabetes mellitus, social isolation, obesity, and excessive alcohol consumption, diverged from the goal of reducing modifiable dementia risk through effective management of modifiable factors.[10,41] Challenges such

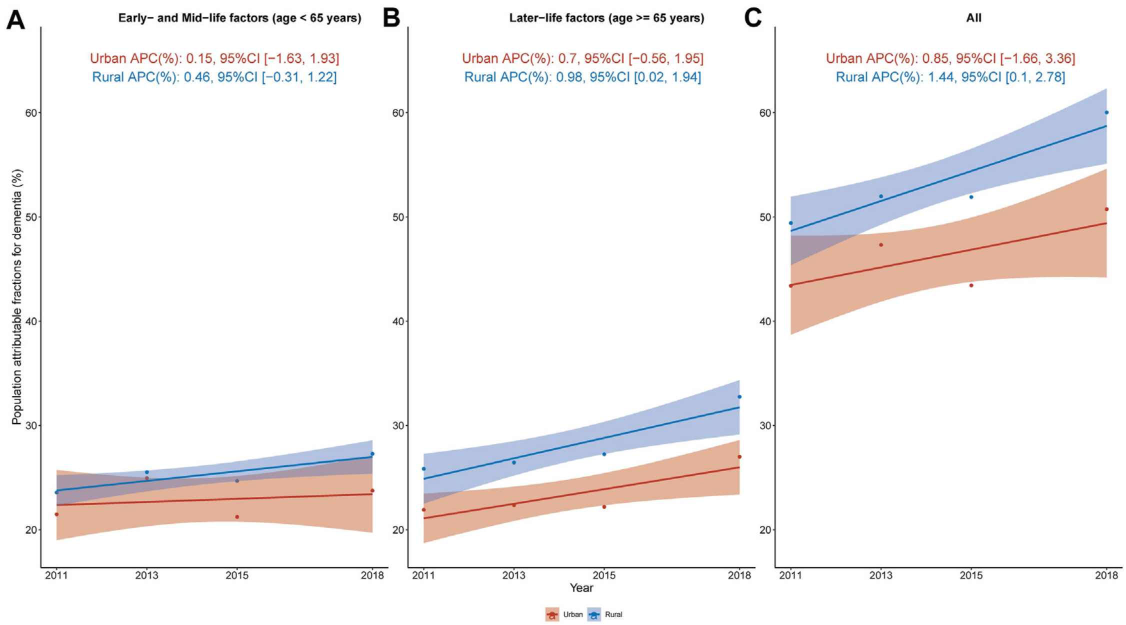

Fig. 7: **Temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018, by rural vs urban.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year.

as late detection of diseases (like diabetes),[56] inadequate prevention (like TBI),[57] limited healthcare access (like depression),[58] unintended consequences of economic development and urbanization (like social isolation),[53] and entrenched lifestyle habits (like physical inactively)[54] may hinder progress. To overcome these challenges, targeted interventions such as improving mental health access and promoting community engagement are essential for effective dementia prevention in China.

The sex-specific analysis of the IW-PAFs of 12 MRFs also provides essential insights for tailored interventions, despite no marked disparities in overall MRF PAF across sexes. Women's higher IW-PAFs for psychosocial factors such as TBI, social isolation, and depression underline the sex-specific risk profiles in new-onset dementia. This sex disparity necessitates focused prevention strategies for women in China. These psychosocial factors are often linked to sex roles and societal expectations, particularly in ageing females who might experience increased social isolation due to life events like widowhood or caregiving responsibilities. This is supported by research, such as Santini et al. (2020), which observed a higher incidence of loneliness among older women.[59] Conversely, men showed higher IW-PAFs for risk behaviours like smoking and alcohol consumption, reflecting prevailing sex norms.[60] The significant reduction in hearing loss among men, not mirrored in women, points to successful sex-specific interventions or better healthcare access for men, but also highlights a potential gap in addressing this issue in women. This suggests the need for more focused efforts in identifying and treating hearing loss in women, possibly requiring different strategies than those for men to ensure equitable healthcare outcomes.

Our SES-based analysis highlights a disproportionate burden of modifiable dementia risk factors on low-income groups in China. This socio-economic disparity emphasizes the urgent need for targeted preventive measures in low-income communities. The pronounced impact of later-life factors, notably depression and social isolation, largely drives this socio-economic disparity in dementia risk. Factors such as low education, TBI, and hearing loss, though less dominant, still contribute to this SES disparity. This pattern is consistent with existing research linking socio-economic status with health outcomes, including dementia risk.[61] Our study contributes to this body of knowledge by offering a nuanced view of how each risk factor contributes to SES-related disparities and helps prioritize interventions and customize approaches. For example, the significant increase in obesity and social isolation's IW-PAF in the lowest SES group and the upward trend in diabetes in the highest group call for tailored lifestyle interventions in more affluent demographics.

Our study expands on geographic disparities in modifiable dementia risk factors, revealing a significant rural-urban divide in China. From 2011 to 2018, this gap

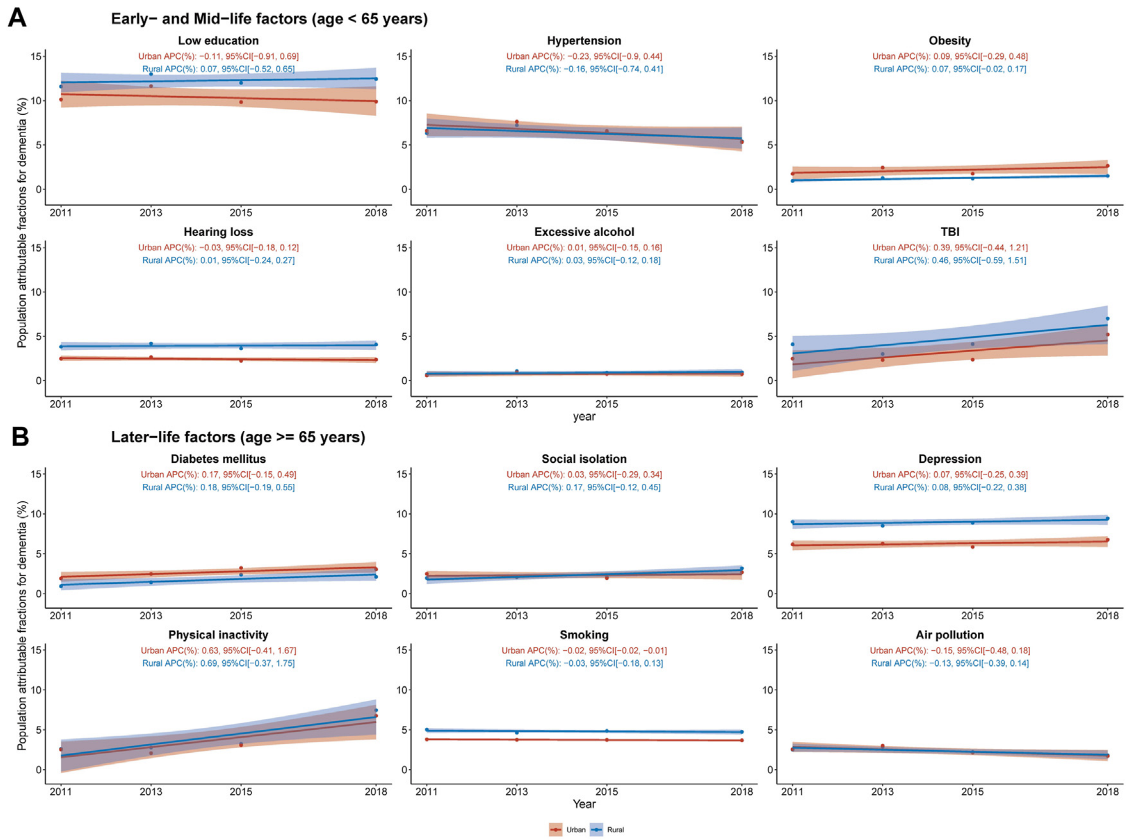

Fig. 8: **Temporal trends in population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018, by rural-urban and risk factor.** Average percentage change (APC) was used to quantify the temporal trend in population attributable fraction (PAF, as %), extracted from linear regression with PAF as the outcome and continuous form of the year as the predictor. The APC indicates the extent to which the percentage points of PAF vary with each passing year. The same figure but with the range of the Y-axis adjusted accordingly based on the observations for each factor, can be found in Supplementary Fig. S4.

has notably widened, challenging the ideal of equitable healthcare access envisioned in public health principles. The higher overall MRF PAF in rural areas was mainly driven by later-life factors like depression, diabetes, and smoking, as well as early-to mid-life factors such as low education, hearing loss, and TBI. This rural-urban disparity is likely a consequence of long-standing imbalances in economic, health, and educational resources, compounded by factors like limited healthcare access and lower health literacy in rural settings.[62,63] Our provincial analysis echoes these geographic discrepancies, with lower overall PAFs in western and eastern/coastal regions compared to central China, indicating diverse socio-economic, healthcare, and lifestyle factors.[64] Provinces such as Hebei, Hunan, Qinghai, and Sichuan experienced significant increases in overall PAF, reflecting growing regional health and social challenges. Areas like Beijing, Shanghai, and Tianjin showed non-significant declining trends, possibly due to better healthcare and public health interventions. The disparity in modifiable dementia risk across provinces is increasing, which expanded from 25.5% in 2011 to 40.6% in 2018. This highlights an urgent need in provinces with a greater increase in PAF. In these provinces, it is crucial to increase awareness, screening, and mitigating measures for these risk factors. Some provinces show significant improvements in low education, hypertension, and air pollution management. However, others face significant rising challenges with low education, depression, smoking, TBI, hearing loss, diabetes, obesity, and excessive alcohol consumption. These variations call for a regionally nuanced approach to dementia prevention. This approach should integrate socio-economic contexts, healthcare capacities, and local health behaviours.

The implications of our findings on the 12 modifiable risk factors for dementia in China should be considered within a framework that involves a complex interplay of social determinants, biological pathways, and intermediary risk factors. Social determinants such as low educational status may be root causes of dementia and intermediary risk factors like alcohol

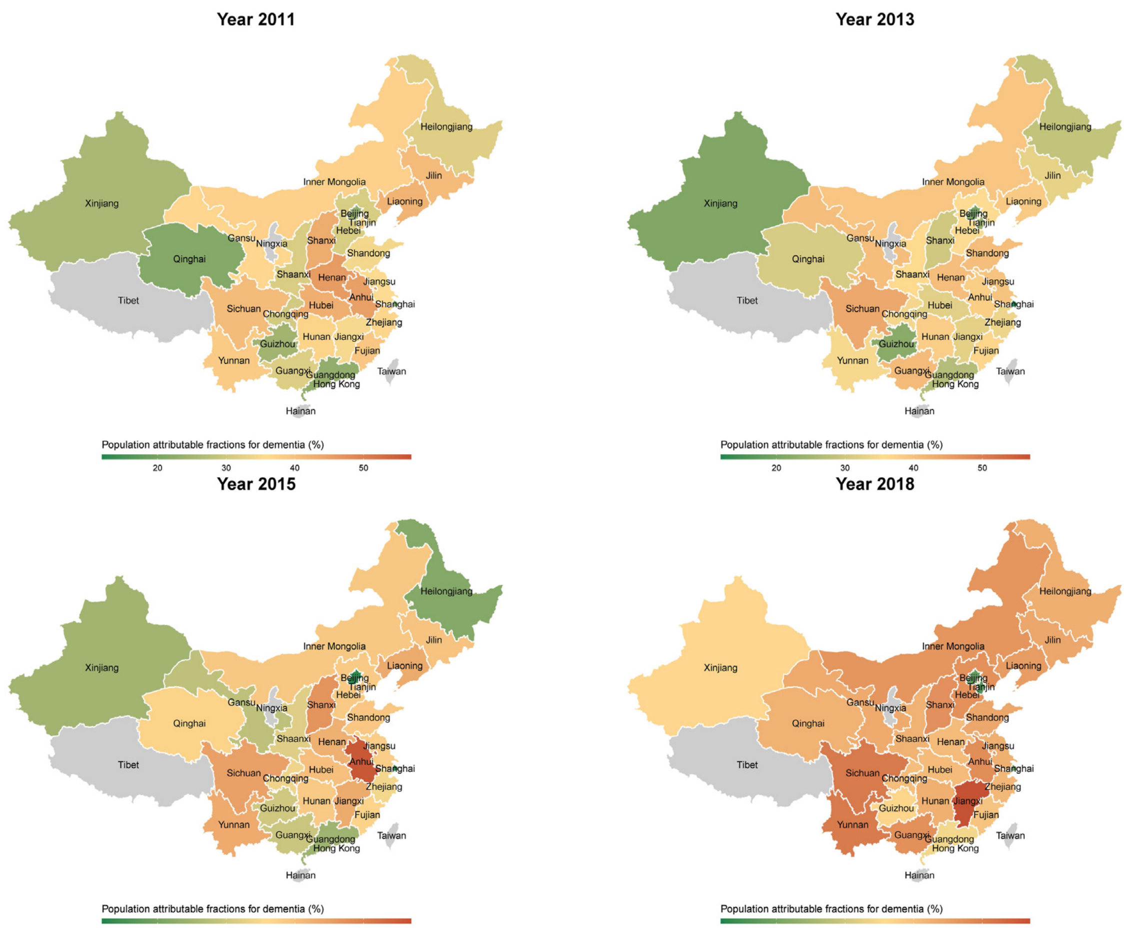

Fig. 9: **Provincial distribution of population attributable fraction of 12 modifiable risk factors for dementia, 2011–2018.** The map only shows areas on land. Grey areas represent unavailable data for the corresponding areas. The table form can be found in Supplementary Table S4.

consumption and hypertension, while diminished socio-economic status may also result from dementia onset. Upstream policies should address the social determinants of health contributing to dementia risk. Given the substantial impact of low education on dementia risk in China, policies aimed at improving access to education, particularly in early life, could have a significant preventive effect. This may involve increasing public funding for education, promoting adult learning programs, reducing educational barriers for disadvantaged populations, and increasing health literacy on dementia among adults. Policies reducing socio-economic disparities and improving living conditions, such as income support programs, affordable housing initiatives, and community development projects, could also address root social determinants of dementia risk.[10,41] Downstream policies should target specific risk factors identified in our study. Traditional public health initiatives promoting physical activity, reducing smoking and alcohol consumption, and encouraging healthy diets should be maintained and could mitigate lifestyle-related risk factors. Given the high IW-PAFs for depression, policies should prioritize access to mental health services. For instance, expanding health insurance coverage, increasing screening programs, and integrating mental health services in primary care. Recent literature also highlights the role of biological determinants and pathways, such as inflammation, in dementia development.[65] Early clinical management of risk factors contributing to inflammation, like obesity, diabetes, and cardiovascular disease, could reduce dementia risk. Lifestyle interventions reducing inflammation, such as exercise and a healthy diet, may also be promising primary prevention strategies.[66]

Strengths and limitations

Our study represents a pioneering and contemporary examination of modifiable risk factors for dementia in China, offering several notable advancements in this field. It is, as far as we are aware, the first study to chart the time-related changes in the MRF PAF for new-onset dementia over an eight-year span in China. This approach provides a detailed and evolving picture of dementia risk factors, allowing for an assessment of the trajectory of these factors and the impact of public health measures over time. Additionally, our analysis breaks new ground by incorporating factors such as sex, SES, and geographic location into the examination of dementia risk. This multifaceted analysis not only highlights the general disparities but also dissects the specific contribution of each risk factor to these disparities. Furthermore, our study is characterized by its rigorous methodology. The RRs used are drawn from the most current meta-analyses, providing a solid evidence base.[10,41] While these RRs do not simultaneously adjust for all other risk factors, we have addressed the interdependence of these factors through the use of communality weights, known for their reliable precision in calculating combined PAFs. Unique to our approach is the sourcing of all analytical elements—prevalence, communalities, and overall PAF—from a consistent data source, enhancing the internal consistency of our findings, an aspect often lacking in similar studies.

When interpreting the results of our study, several limitations must be acknowledged.

First, it is important to note that our study assumes the associations between risk factors and dementia, as reflected in the relative risk estimates, are causal. Consequently, we posit that removing these risk factors would lead to a reduction in dementia incidence, as indicated by the PAFs. However, this assumption should be critically considered as a potential limitation. While the relative risk estimates are derived from current meta-analyses, the causal nature of these associations cannot be definitively established through observational studies alone. Residual confounding factors or reverse causation may influence the observed associations. Furthermore, the PAF approach models an idealistic scenario of complete risk factor elimination, which may not be entirely practical. Reductions in these factors can indeed help delay or prevent dementia onset, potentially lowering its incidence. However, the interpretation of PAFs as direct causal effects should be approached with caution, and further research, particularly intervention studies, is needed to establish causality more conclusively. Further, the PAF model overlooks the possibility of increased dementia incidence due to extended lifespans achieved through risk factor reduction.[8] Therefore, our estimates of incidence reduction might lean towards the optimistic side. It is also plausible that individuals who live longer, having mitigated these risk factors, might have a reduced likelihood of developing dementia, contributing to a lower age-related incidence. This is in line with the observed trend of declining dementia incidence in several countries,[67,68] despite the ageing population. Additionally, the presentation of RRs in PAF studies is typically binary, not reflecting the continuous relationship between risk factor levels and dementia risk.[42] The PAF formula considered the prevalence overall between factors, while the Lancet Commission's estimation of relative risks (RRs) did not. Although the Commission used meta-analysis, a strong evidence-based approach, not all original studies controlled for other risk factors when estimating RRs. This potential overlap between factors was not accounted for, possibly leading to overestimated RR values.

Second, our study encountered limitations stemming from the imprecision in data related to the prevalence of factors such as hearing loss, TBI, and depression. While we relied on self-reported data and survey scales for estimating prevalence, a method consistent with prior research focused on PAF,[7–9] it is crucial to acknowledge that such methodologies can lead to potential inaccuracies, causing both overestimations and underestimations of true prevalence rates.

Third, while our study utilizes the 12 MRFs identified by the Lancet Commission, it may not capture all potential risks, such as healthcare accessibility,[69] and might not fully represent the increased dementia risk in socio-economically disadvantaged groups noted in other research.[9,70]

Fourth, the CHARLS employs a cohort design, focusing on individuals aged 45 and above. This age range captures the population most at risk for dementia, but the CHARLS data only included individuals residing in private households and did not encompass those living in residential care facilities. This may lead to an underestimation of our PAFs estimations. Studies have shown that individuals in residential care facilities may have a different profile of risk factors, such as higher rates of physical inactivity, depression, and social isolation, than in community-dwelling populations.[71,72] However, the proportion of older adults living in residential care facilities in China is relatively low (about 3%).[73] Therefore, while the exclusion of this population may lead to an underestimation of dementia prevalence and PAFs, the magnitude of this effect may be relatively small in the Chinese context.

Fifth, attrition in CHARLS can potentially affect the representativeness of MRFs if the attrition is related to these factors. Considering the typically higher mortality rates observed among individuals with lower SES, those residing in the western regions of China, and males,[74,75] it is plausible that the PAF disparities identified in our study may be underestimated compared to what would be observed in a newly conducted cross-sectional survey. Despite CHARLS providing sampling weights to ensure representativeness in each wave and implementing refreshment samples to maintain population representativeness, the effectiveness of these measures in maintaining representativeness over time, particularly with respect to the modifiable risk factors, warrants further investigation.

Sixth, a relatively higher proportion of missing values was observed among participants with the lowest wealth status, those living in rural areas, and those with low education levels. Consequently, our PAF estimations for those in deprived situations may be underestimated. Nevertheless, the sensitivity analysis without imputation also confirmed the consistency of our main findings.

Seventh, a marked increase in the proportion of individuals aged 65 and above was observed in the CHARLS dataset, from 29.8% in 2015 to 37.4% in 2018, also exceeding the 31.7% reported in the 2020 census. Although the CHARLS refreshes its cohort in each wave, Supplementary Table S1 showed fewer new participants were recruited in 2018. This suggests that the cohort is ageing rather than reflecting population changes, potentially leading to overestimated PAF estimates for 2018 due to the higher representation of older individuals compared to the general population.

Finally, the data used in our study precedes the COVID-19 pandemic,[28,76] an event that has notably impacted older adults and those with cognitive impairments. The pandemic's influence on new-onset dementia risk factors and incidence remains outside the scope of our data and is an area for future research.

Generalizability, implications, and conclusions

Over an eight-year period, our extensive longitudinal study has shed new light on the changing dynamics of modifiable dementia risk factors, considering the influence of sex, socio-economic status (SES), and geographical differences. Our approach, grounded in the latest meta-analyses, lends significant internal coherence to our findings. Yet, these insights also pave the way for future inquiries, particularly in examining the impact of the COVID-19 pandemic on these risk factors. Our research has profound implications for practical applications, despite certain limitations. It emphasizes the need for dementia prevention strategies that are specifically tailored to address the diverse and unique challenges faced by various demographic groups.

Contributors

SC contributed to the concept and study design. SC conducted the analysis. SC and LL interpreted the results. SC drafted the manuscript. XC, XH, HF, GGL, and LL made critical revisions to the manuscript for important intellectual content. All authors edited and approved the final manuscript.

Data sharing statement

The data are publicly available and can be accessed here (https://charls.pku.edu.cn/en/).

Editor note

The Lancet Group takes a neutral position with respect to territorial claims in published maps and institutional affiliations.

Declaration of interests

SC and all other others declare no conflict of interest with this work.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.lanwpc.2024.101106.

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